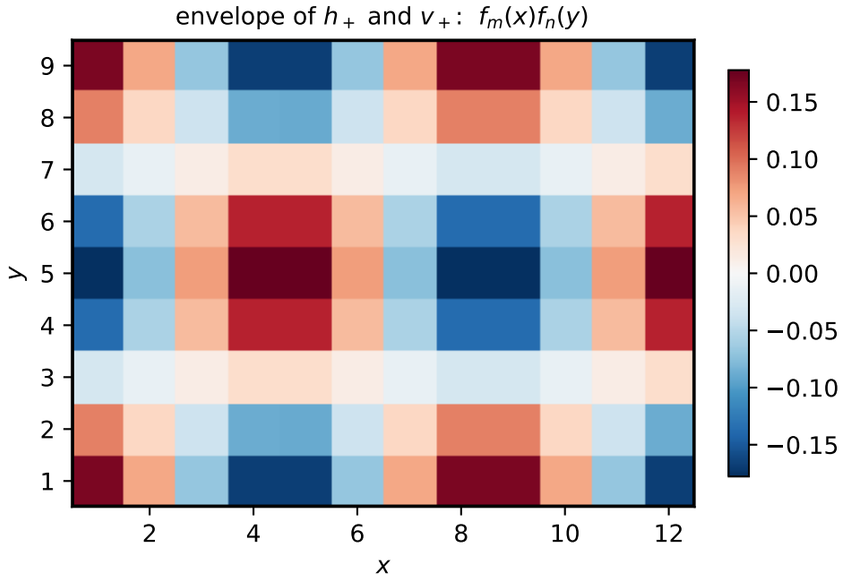

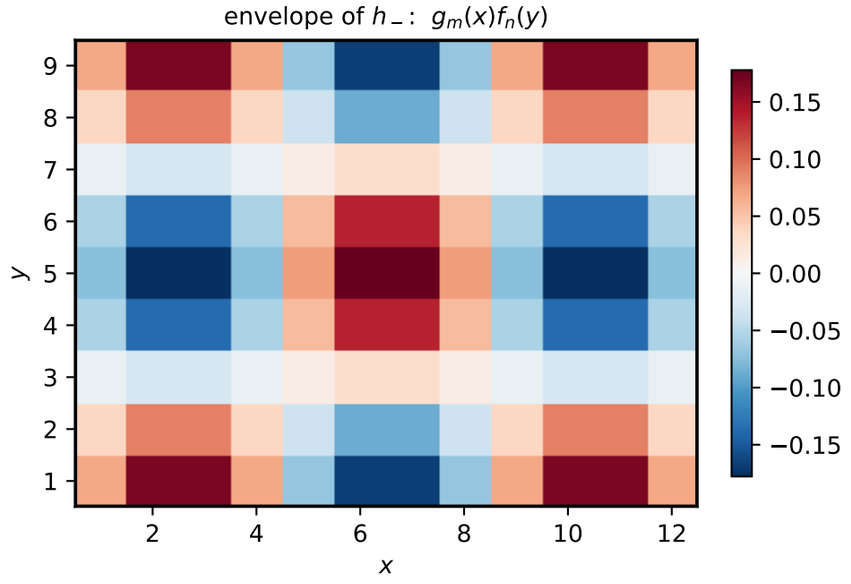

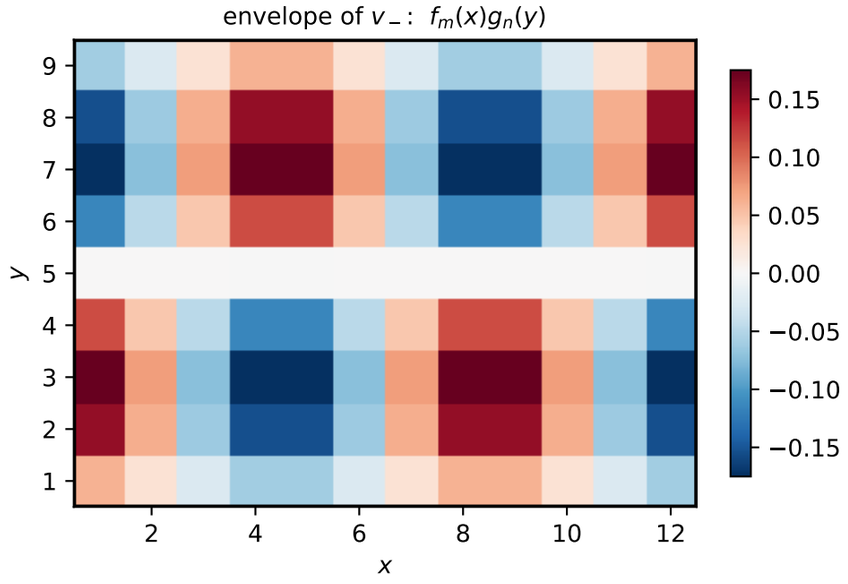

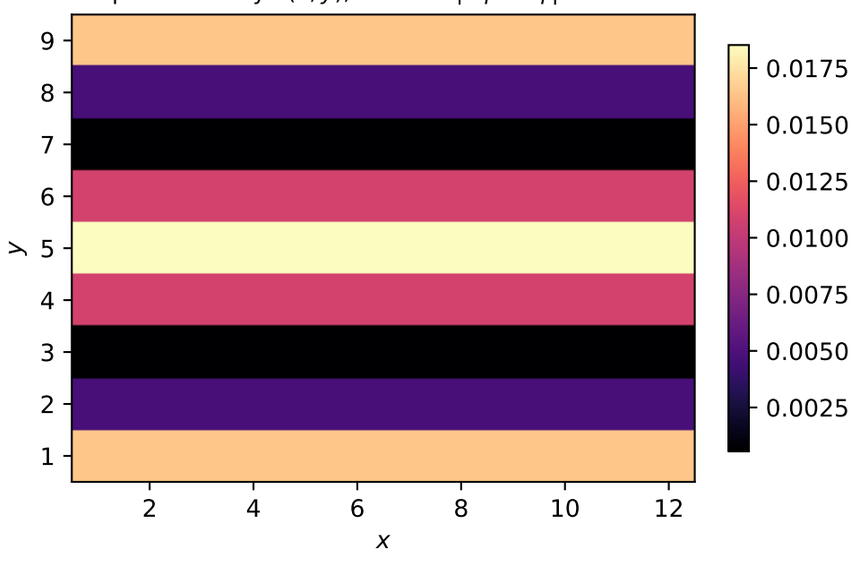

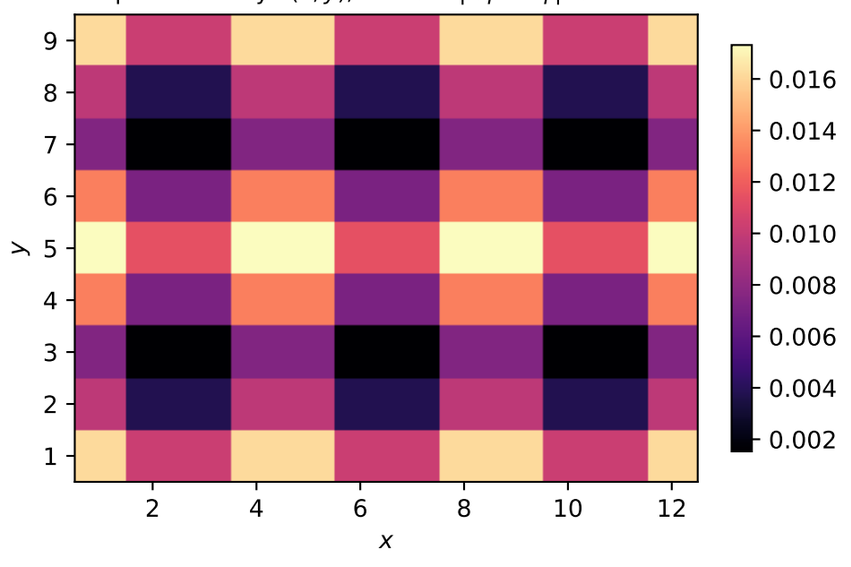

Rectangle 12×9**, mode** $(m, n) = (3, 2)$

Top: the three distinct envelopes of the sector $\mathcal{K}_{mn}$ (red/blue = sign). The walls are the black frame; cosines have zero slope at the wall, sines change sign across the mirror plane.

Bottom left: with $C = I$ the sector splits into an x-pair (a, c) and a y-pair (b, d). The eigenstate $(a - ic)/\sqrt{2}$ runs around the row cycle: its density is uniform along x ($f_m^2 + g_m^2$ is constant) and keeps only the y standing wave.

Bottom middle: Grover couples h_+ with v_+, so its eigenstates mix all four channels and show a genuine two-dimensional standing-wave pattern.